

Hand and finger injuries

with HAMISH ANDERSON and GREGORY HOY

Star midfielder Daniel Kerr says he's willing to risk permanent damage to his injured finger if it means helping the Eagles reach and win another Grand Final. West Coast's bid for a third straight grand final, and back-to-back premierships, was dealt a shattering blow last week when Kerr had to undergo surgery on a ruptured tendon in the ring finger of his left hand.

Superfooty AFL, 29 August 2007

Sports-related injuries account for up to 15% of all hand injuries seen in accident and emergency departments.¹ Some hand injuries are potentially serious and require immobilisation, precise splinting and/or surgery. Finger injuries are often neglected by athletes in the expectation that they will resolve spontaneously. Athletes often present too late for effective treatment, especially with ligamentous injuries and joint instability.

The importance of early assessment and management must be stressed so that long-term deformity and functional impairment can be avoided. Many hand and finger injuries require specific rehabilitation and appropriate protection upon resumption of sports. Joints in this region do not respond well to immobilisation, which often exacerbates oedema and stiffness. Hence, surgical stabilisation and protected stabilisation with active motion splints are becoming commonplace for sports-related hand injuries.

As in all injury management, return to occupation is important, and if the occupation is sport, then more aggressive management options are often employed.

The sport-specific causes of pain in the hand are shown in Table 27.1. The anatomy of this area is demonstrated in Figure 27.1.

CLINICAL APPROACH

History

The mechanism of injury is the most important component of the history of acute hand injuries. A direct, severe blow to the fingers may result in a fracture, whereas a blow to the point of the finger may produce an interphalangeal

dislocation, joint sprain, or long flexor or extensor tendon avulsion.

A punching injury often results in a fracture at the base of the first metacarpal or to the neck of one of the other metacarpals (usually the fifth), or carpo-metacarpal (CMC) joint instability. An avulsion of the flexor digitorum profundus tendon, most commonly to the fourth finger, is suggested by a history of a patient grabbing an opponent's clothing while attempting a tackle.

Lack of active motion is an important warning sign of significant injury. Associated features such as an audible crack, degree of pain, swelling, bruising and loss of function should also be noted.

Patient outcome measures are also useful, particularly in determining levels of function in chronic cases. The Michigan Hand Outcomes Questionnaire and the Disabilities of the Arm, Shoulder and Hand are widely used evaluations³⁻⁵ (Table 27.2).

Physical examination

At the outset of examination, examine the posture of the hand, the protection of movement and the deformities compared to the other (hopefully) uninjured hand. Ask the athlete to make a fist and open, and to oppose the thumb to each finger on each side.

Carefully palpate the bones and soft tissues of the hand and fingers, looking for tenderness. The examiner should always be conscious of what structure is being palpated at any particular time. The joints should be examined to determine active and passive range of movement

Table 27.1 Sport-specific hand conditions

Injury	Common name	Common sports	Management
Flexor digitorum profundus avulsion	Jersey finger	Tackling sports: rugby, all football codes	Early: Surgical repair Late: (>6 weeks) two-stage reconstruction
Flexor tendon pulley rupture (especially A2 and A4)	Climber's finger	Rock climbing and mountaineering	Taping (preventative) Splints (acute) or surgical pulley reconstruction
Sagittal band rupture MCPs 2,3,4	Boxer's knuckle	Boxing	Sagittal band reconstruction
Ulnar digital nerve neuroma	Bowler's thumb	10-pin bowling	Padding to thumb hole Surgical neurolysis
Ulnar collateral ligament rupture MCP thumb	Skier's thumb (gamekeeper's thumb)	Skiing, football ('thumb plant')	Splint if incomplete Surgical reconstruction if Stener lesion
Radial collateral ligament tear MCP thumb	AFL thumb	Australian rules football	Tape +/- splint to finish season Surgical reconstruction
CMC index/middle subluxation and carpal bossing	Boxer's hand	Boxing, martial arts	Splints (for competition without surgery) Resect bossing (short term) CMC fusion
Hook of hamate fracture		Batting and clubbing sports	Early: Protect Late: Excision of hook of hamate
Stress fracture scaphoid		High board diving	Splint or cast for 6 weeks
Stress reaction lunate (differential diagnosis: Stage 0 Kienbocks disease)		Golf, tennis	Splint or cast for 6 weeks Change technique

and stability. Stability should be tested both in an antero-posterior direction and with ulnar and radial deviation to assess the collateral ligaments. Do not forget that a depressed fragment of bone in a joint may produce a false positive test when stretching the redundant collateral ligament. The cause of any loss of active range of movement should be carefully assessed and not presumed to be due to swelling. Normal range of motion for the second to fifth digits is approximately 80° of flexion at the distal interphalangeal, 100° of flexion at the proximal interphalangeal and 90° of flexion at the metacarpophalangeal (MCP) joint. A common injury site that can be overlooked is the volar plate, a thick fibrocartilagenous tissue that reinforces the phalangeal joints on the palmar or volar surface.

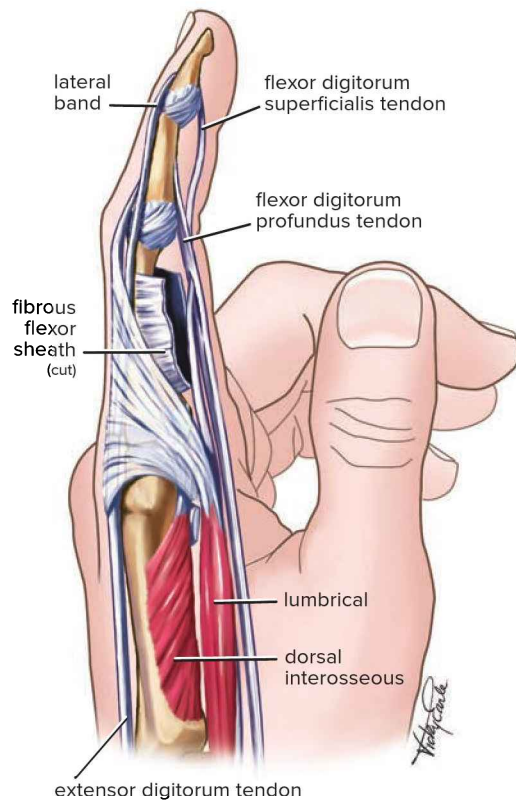
The extensor tendons of the hand are often divided into six compartments. At the wrist on the dorsal side of the hand, the tendons are encased in synovial sheaths as they pass under the extensor retinaculum (Fig. 26.1). When palpating in the most radial compartment, also known as compartment one, the examiner identifies abductor pollicis longus and extensor pollicis brevis, the

tissues involved in de Quervain's tenosynovitis. Lister's tubercle is located on the dorsal surface of the distal end of the radius. The extensor pollicis longus angles sharply around this bony prominence and it can damage or even rupture the tendon after a serious wrist fracture.

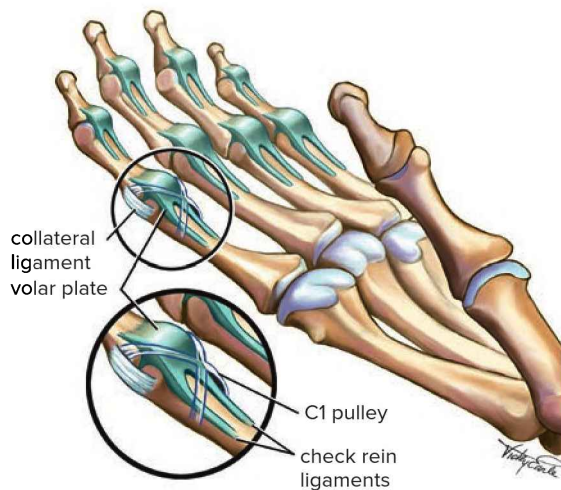
The anatomical snuffbox is composed of the extensor pollicis longus and brevis and the abductor pollicis longus. The floor of the snuffbox is the scaphoid and scaphotrapezium joint. Clinically this is a significant region for several reasons. Tenderness may suggest scaphoid fracture. The deep branch of the radial arterial passes through, as well as the superficial branch of the radial nerve. Consequently, if a cast or splint is applied too tightly, it can lead to numbness in the thumb.

Examination of the fingers and hand involves:

1. Observation and sensation testing as per the wrist (Chapter 26). Special note should be made of the hand arches and any deformities at the proximal or distal interphalangeal joints.
 - a. hand at rest
 - b. hand with clenched fist (Fig. 27.2a)



(a)



(b)

Figure 27.1 Anatomy of the metacarpals and fingers

(a) Muscles and tendons (b) The volar plate

2. Active movements—fingers (all joints)

- flexion
- extension
- abduction
- adduction

3. Active movements—thumb

- flexion
- extension
- opposition (Fig. 27.2b)

4. Resisted movements (to test tendon integrity)

- flexor digitorum profundus (Fig. 27.2c)
- flexor digitorum superficialis (Fig. 27.2d)
- extensor tendon (Fig. 27.2e)

5. Special test

- ulnar collateral ligament of the first MCP joint (Fig. 27.2f)
- IP joint collateral ligaments

Sensitivity and specificity of testing the stability of the ulna collateral ligament of the first MCP joint has been demonstrated to improve significantly to 100% and 87.5% post injection of local anaesthetic to the joint. The accessory collateral ligament at the proximal interphalangeal (PIP) joint is tested for lateral laxity in extension, whereas the cord collateral ligament is stress tested with the PIP joint flexed to 45°.²



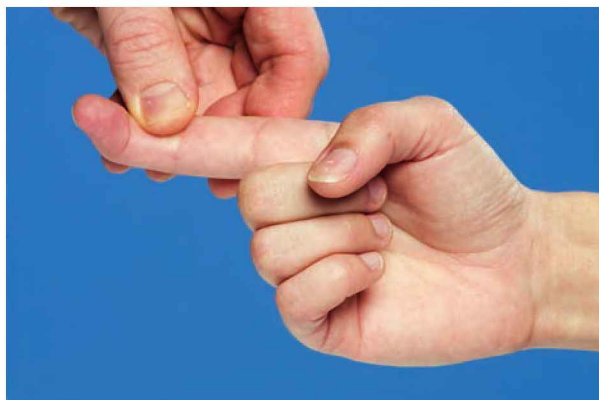
(a)



(b)

Figure 27.2 (a) Attitude of hand with clenched fist

(b) Thumb movement—opposition



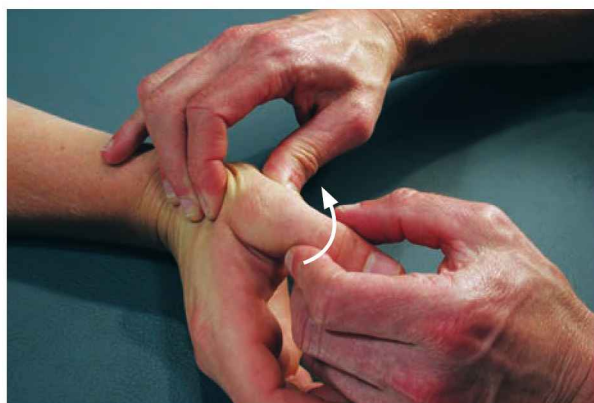
(c)



(e)



(d)



(f)

Figure 27.2 (cont.) (c) Tendon integrity—flexor digitorum profundus. The patient flexes the DIP joint with the PIP joint held in extension (d) Tendon integrity—flexor digitorum superficialis. The patient flexes the PIP joint with the other DIP joint held in extension (e) Tendon integrity—extensor tendon. The patient extends the PIP joint with the MCP joint in extension (f) Special test—the ulnar collateral ligament of the thumb is tested with 10° of flexion at the first MCP joint

Key outcome measures

Refer to Table 27.2 (Refer also to Chapter 16 Patient reported outcome measures in sports medicine).

Investigations

Routine radiographs of the hand include the PA, oblique and lateral views. Ideally, 'dislocations' need to be radiographed before reduction to exclude fracture and after reduction to confirm relocation. Even when pre-reduction radiographs are not performed because reduction has occurred on the field, post reduction films should be obtained after the game. Care should be taken with lateral views to isolate the affected finger to avoid bony overlap.

The use of more sophisticated investigation techniques is now more routine. CT scans show subtle avulsions and 3D CT scans are useful for understanding fracture patterns and methods of reduction. Single-photon emission

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All traumatic finger injuries should be X-rayed.

computed tomography (SPECT) imaging combined with CT scans can show hot spots and stress reactions in bone as well as excluding bone lesions in some circumstances. MRI scanning is now very common for ligament and tendon injuries to grade and assist surgical decision making. Ultrasound is still widely indicated in dynamic imaging such as with subluxing tendons, the Stener lesion in the thumb, and siting tendon-ends in ruptures and lacerations.

PRINCIPLES OF TREATMENT

The functional hand requires mobility, stability, sensitivity and freedom from pain. It may be necessary to obtain

Table 27.2 Patient-reported outcome measures (PROMs) for hand and finger injuries and pain

Michigan Hand Outcomes questionnaire
1. Hand-specific assessment covering function, ADLs, pain, work performance, aesthetics and satisfaction
2. x items: 37
3. Score range x–y
4. Test–retest reliability: >0.81 ³
5. Validity: High validity demonstrated
6. MDC95%: Varies according to diagnosis
7. MCID: Varies according to diagnosis
8. Available online: Yes
Disability of the Arm, Shoulder and Hand questionnaire (DASH and Quick DASH)
1. Measures physical function, symptoms and quality of life in people with musculoskeletal disorders of the upper limb
2. x items: DASH 30; Quick DASH 11
3. Score range x–y: 0 (no impairment)—100 (poor)
4. Test–retest reliability: DASH >0.96; ⁴ QuickDASH >0.91 ⁵
5. Validity: DASH >0.70; ⁴ QuickDASH > 0.70 ⁶
6. MDC95%: DASH 12.75%; ⁴ QuickDASH 11.2% ⁵
7. MCID: 10.83%; ⁴ 8% ⁵
8. Available online: DASH and QuickDASH

stability by surgical methods. However, conservative rehabilitation is essential to regain mobility and long-term freedom from pain. Treatment and rehabilitation of hand injuries is complex. As the hand is unforgiving of mismanagement, practitioners who do not see hand injuries regularly should ideally refer patients to an experienced hand therapist, or at least obtain advice while managing the patient.⁷

Inflammation and swelling are obvious in the hand and fingers. During the inflammatory phase, the therapist must aim to reduce oedema and monitor progress by signs of redness, heat and increased pain. During the regenerative phase (characterised by proliferation of scar tissue), the therapist can use supportive splints and active exercises to maintain range of motion. During remodelling, it is appropriate to use dynamic and serial splints, and active and active assisted exercises, in addition to heat, stretching and electrotherapeutic modalities.⁸

Oedema control

Oedema can be controlled through splinting, compression, ice, elevation and electrotherapeutic modalities (Chapter 17). Splinting needs to be in the intrinsic plus position, with the wrist in 30° of dorsiflexion, the MCP joints flexed

to 70° and the PIP joint extended to 0° with the thumb abducted (Fig. 27.3). Splints are periodically removed to allow exercise. Fist-making exercises are used to maintain joint movement and to help remove oedema. During exercise, the hand should be elevated. Short, frequent exercise periods are optimal.



Figure 27.3 The ‘intrinsic plus’ position for splinting of hand injuries: 30° of wrist dorsiflexion, 70° of MCP flexion and minimal PIP flexion

Compression in the hand can be achieved with a compression glove and by using a Coban™ elastic bandage: a 2.5 cm size is appropriate for fingers. If applicable, active tendon gliding and range of motion exercises in combination with elevation can assist in the reduction of swelling. Electrotherapeutic modalities can be useful in the control of oedema.

Exercises

Exercises may be active, active assisted or resisted. Tendon lacerations or ruptures are generally treated by protocols determined by the surgical technique and preferences of the treating surgeon.

Exercise prescription for other injuries includes:

- blocking exercises, which isolate an injured joint or muscle by immobilising neighbouring joints with the other hand
- composite flexion exercises (e.g. tendon gliding exercises)
- extension exercises
- active assisted exercises in which the patient takes a joint through the available range of motion and then the therapist assists in gaining slightly greater range with overpressure.

Taping and splinting

The most commonly used method of taping is 'buddy taping' (Fig. 27.4). Its role is to provide a vehicle for active assisted exercise. The uninjured digit provides additional stability and encourages full range of motion.

An injury to the finger can often be managed successfully with a conservative approach using the protection of some form of splint. Thin thermoplastic—1.6 mm or less is ideal—can be remoulded to accommodate changes in swelling or protocol. Finger sleeves (Fig. 27.5a) can be made of neoprene or Lycra® to manage oedema, pain or chronic



Figure 27.4 Buddy taping of fingers

flexion deformity, and both can be reinforced with thin thermoplastic if extra stability is required.⁹

Splints can have padding added and be taped on for protection during sporting activity. The goal is to provide stability, but not at the expense of limiting movement and impeding performance.¹⁰ The use of splints during a game is entirely dependent upon the athlete, his or her sport and the rules governing the sport at that level.

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1.6 mm thermoplastic can be softened sufficiently in a hot cup of water to make a splint on the sidelines.

In the acute phase of injury, static splints are used to reduce oedema. Dynamic splints can also be used in the repair phase of injury to provide some force along joints and encourage increased range of motion (Fig. 27.5b and c). With dynamic splinting, the splint should be worn with less tension for a longer period.^{9, 10}

METACARPAL FRACTURES

Fracture of the base of the first metacarpal

Fractures of the base of the first metacarpal commonly occur as a result of a punch connecting with a hard object, such as an opponent's head, or a fall on the abducted thumb. There are two main types of fracture: (1) the extra-articular transverse fracture of the base of the first metacarpal about 1 cm distal to the joint (Fig. 27.6a) and (2) a Bennett's fracture/dislocation (or Rolando comminuted fracture/dislocation) of the first carpo-metacarpal joint (Fig. 27.6b).



(a)

Figure 27.5 Splints used to treat fixed flexion deformity
(a) Neoprene finger sleeve provides gentle extension also addresses chronic oedema



(b)



(c)

Figure 27.5 (cont.) (b) Static progressive (also known as a 'belly gutter') splint. This simple finger gutter splint has a 'belly' underneath the PIP joint. It is usually put on half an hour before bed with a gentle tension on the central strap. This provides a progressive and controlled end-range stretch into extension overnight (c) A commercial dynamic (spring) finger-based PIP extension splint. This splint is only suitable for contractures of less than 40° and does permit active flexion

The transverse fracture near the base of the first metacarpal results in the thumb lying flexed across the palm. Reduction of this fracture involves extension of the distal segment of the metacarpal. This fracture can be immobilised in a short arm spica cast, but now more commonly internal fixation allows both anatomical reduction and early active movement. This produces prompt return to activity, and less delay in recovery from loss of position. The often-quoted 'surgery only as a last resort' is less relevant today with better surgical training, better imaging and modern surgical implants.

A Bennett's fracture dislocation of the first carpo-metacarpal joint occurs as a result of axial compression



(a)



(b)

Figure 27.6 Fractures of the base of the first metacarpal (arrows) (a) Healing transverse fracture (b) Bennett's fracture

when the first metacarpal is driven proximally, shearing off its base. A small medial fragment of the metacarpal remains attached to the strong volar ligament and the main shaft of the metacarpal is pulled proximally by the unopposed pull of the abductor pollicis longus muscle. This injury should be referred to a hand surgeon.

Treatment in recreational athletes requires closed reduction and percutaneous Kirschner wire fixation together with cast immobilisation for 4–6 weeks. Upon removal of the cast, mobilisation of the surrounding joints is required and, if early return to sport is required, a protective device should be worn.

Professional athletes will have anatomical reduction, plate fixation and early active movement. Patients not engaging in contact sport find soft neoprene braces (Fig. 27.7) supportive and comfortable after a Bennett's fracture and other common hand injuries. These are not replacements for firmer splints and braces that might be needed when trauma can be anticipated.

Fractures of the other metacarpals

Fractures of the second to fifth metacarpals may also occur as the result of a punch, contact with ball and ground, and direct hits at hockey, cricket, baseball, etc. When seen in the fourth and fifth metacarpals, it is referred to as a 'boxer's fracture'. The fracture is usually accompanied by considerable flexion deformity of the distal fragment which is countered by the extra dorsiflexion available at the MCP joint to make up for this deformity. This results in surprisingly little functional disability (Figs 27.8 and 27.9).

The acceptable angulation for fractures of the neck of the fourth and fifth metacarpals is up to 30°, as long as there is little rotational deformity. Up to 10° angulation is acceptable in the second and third metacarpals. However,



Figure 27.7 Soft neoprene splints can provide support during rehabilitation after a hand injury

prominence of the metacarpal head in the palm of the hand may be a problem for tennis players and athletes who require a firm grip.

Treatment involves splinting or casting in a position of 70° of flexion of the MCP joints to prevent shortening of the collateral ligament and subsequent stiffness (Fig. 27.10a). Check that this position does not displace the fracture. The splint may be removed after 2–3 weeks and non-contact sport resumed immediately with

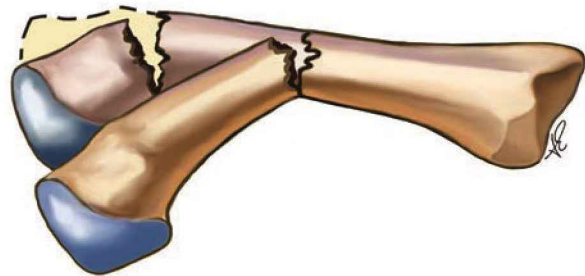


Figure 27.8 In metacarpal fractures, the more proximal the fracture, the more the knuckle will drop

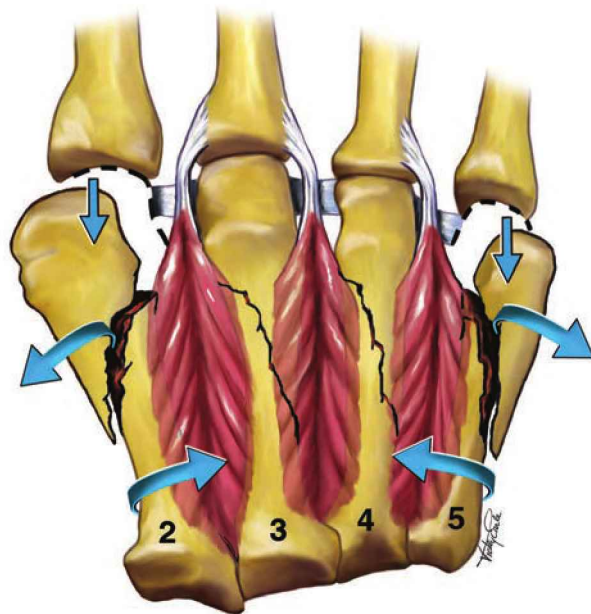


Figure 27.9 Spiral fractures of the second and fifth metacarpals (border digits) are more unstable than those of the third and fourth metacarpals as only one side of the border metacarpals is supported by the strong, deep transverse intercarpal ligaments

protection (Fig. 27.10b). Unprotected return to contact sport is more controversial, and recently suggestions of aggressive return to play 1–2 weeks after strong plating in professional sports have been put forward.

Shortened, rotated and intra-articular fractures of the metacarpals require anatomical reduction and fixation. In

displaced fractures, this usually involves plating from the dorsal or lateral aspect to correct angulation and rotation. Check for rotation of finger fractures clinically (as subtle rotation causes 'crossing' of metacarpals and affects grip) and not by using radiographs. Undisplaced fractures can be immobilised in a gutter splint with flexion of the MCP joint.¹¹



(a)



(b)

Figure 27.10 Splints used in the management of fracture of the fifth metacarpal (a) The treatment splint should position the MCP joints of the 4th and 5th fingers in 70° of flexion, leaving the IP joints free to move (b) Protective splint used for early return to sport when appropriate. Note the bubble over the fracture site to accommodate swelling

FRACTURES OF PHALANXES

Proximal phalanx fractures

Fractures of the proximal phalanx may lead to functional impairment due to the extensor and flexor tendons coming into contact with callus and exposed bone, and due to imbalance of the extensor expansion with flexors creating 'Z' deformities through the PIP and distal interphalangeal (DIP) joints in hypermobile individuals.

Undisplaced fractures require weekly radiographs to ensure movement has not occurred. If further stability is needed, the adjacent finger can be buddy taped, but it is more common now to pin, wire or plate these fractures to regain stability and allow hand therapy to achieve better early results. Splints are removed after a maximum 3–4 weeks but ideally earlier to improve function. Unstable fractures require urgent surgical referral.

Rotational deformity of phalangeal fractures may not be obvious in extension so the fingers should be examined end on with PIP and DIP flexion to reveal any deformity present. All rotated fractures need open reduction and possibly internal fixation.

Middle phalanx fractures

Fractures of the middle phalanx involve hard cortical bone. Generally oblique or transverse, these fractures heal slowly. The central slip of the extensor tendon attaches dorsally to the base of the bone and the flexor digitorum inserts on the volar surface more distally. Thus, fractures distal to the flexor tendon attachment show flexion of the proximal fragment and extension of the distal fragment.

Stable fractures are immobilised in a splint for 3 weeks (70° of MCP joint flexion and 0° of PIP joint flexion). When the splint is removed, range of motion exercises are begun.

Unstable fractures, or intra-articular fractures involving more than 25% of the PIP joint surface, require open reduction and internal fixation. Small-calibre Kirschner wires or screws are used and range of motion exercises are begun as soon as fixation is considered to be stable in order to avoid stiffness.

Volar plate avulsion fractures can occur at the PIP joint following a hyperextension injury (and often a joint subluxation or dislocation). This injury is very common,



Figure 27.11 Radiograph confirms the subtle nature of a volar plate avulsion

and usually ignored owing to an unawareness of the potential consequences. The typical anatomy is shown above in Figure 27.1b and the radiographic appearance is shown in Figure 27.11.

Non-randomised controlled trials have compared early mobilisation with splinting and found that mobilisation has led to good functional outcomes,¹² although extension blocking helps prevent hyperextension and recurrent dislocations. Often flexion deficits occur when the volar fragment and scarred volar plate are too thick and interfere with tight grip. Excision of the fragment and volar plate arthroplasty (VPA) helps eliminate this problem.

Distal phalanx fractures

Fractures of the distal phalanx are usually caused by crushing injuries, such as fingers being jammed between a fast-moving ball and a stick or bat. They are usually non-displaced.

Often a splint and compression dressing will provide adequate treatment for non-displaced fractures. Much of the pain associated with these fractures can be due to subungual haematoma. Significant subungual haematoma requires nail bed exploration and excision as the nail bed is often disrupted. In this case surgical repair may be required to prevent future nail deformity. This represents a compound fracture and should be treated as such.

Perforation of the nail to drain a subungual haematoma is a relative contraindication in this instance as it

converts a closed fracture to a compound fracture and requires careful prevention or prophylaxis of infection. A percutaneous K-wire from the tip down across fracture and DIP joint allows early function but not contact sport.

PRACTICE PEARL

Most distal phalangeal fractures heal in 4–6 weeks.

DISLOCATIONS OF THE CARPO-METACARPAL JOINTS

Most carpo-metacarpal (CMC) subluxations (with or without fractures of the carpal bones) occur with hitting, such as martial arts, or impact with the ground. Plain radiograph demonstrates the incongruence and the subluxation (Fig. 27.12). While splints can protect temporarily, the best treatment of this injury is reduction and pinning (acutely) or fusion of the CMC joints involved (usually index and middle finger metacarpals to trapezoid and capitate) if chronic.¹³

Dorsal dislocation of the MCP joints of the fingers is uncommon and usually occurs in the index finger or thumb. It has been called the 'irreducible dislocation' because the metacarpal head is pushed through the volar plate of the MCP joint and caught between the lumbrical and long flexor tendons with a buttonholing effect. Suspect this injury when examination reveals hyperextension of the involved MCP joint with ulnar deviation of the finger overlapping the adjacent finger.

An attempt to reduce the dislocation may be made by increasing the deformity and pushing the proximal phalanx through the tear in the volar plate. However, open reduction is sometimes required. The MCP joint is



Figure 27.12 Radiograph of CMC. Sagittal view showing dorsal subluxation of metacarpal (arrow)

usually stable after reduction in 30° of MCP flexion and early movement can be commenced in a dorsal splint allowing full flexion but preventing the last 30° of MCP joint extension. The immobilisation is maintained for 5–6 weeks. Associated osteochondral fractures require open reduction and internal fixation, and the collateral ligaments should be checked (even by MRI if required) to ensure accurate reduction.

DISLOCATIONS OF THE FINGER JOINTS

Dislocations of the PIP joint

Dorsal PIP joint dislocations are the most common hand dislocation. They usually result from a hyperextension stress with some degree of longitudinal compression such as may occur in ball sports. This may produce disruption of both the volar plate and at least one collateral ligament. Ideally, radiographs should precede reduction to confirm the diagnosis and exclude an associated fracture. In practice, reduction often occurs on or beside the playing area so it is vital that radiography be performed afterwards. If a fracture is present, the stability of the joint must be tested.

If the joint is stable post-reduction, a splint is not necessary and buddy taping to prevent hyperextension with active movement is recommended. If the joint is unstable, then the joint should be splinted in flexion beyond the point of instability and gradually extended to neutral over a period of 3–4 weeks¹⁴ (Fig. 27.13).

Fixed flexion deformities are common following this injury in spite of treatment; however, if left untreated, a hyperextension deformity and instability may develop. Swelling around the joint can last for several months and should be managed using an elastic pressure bandage or neoprene sleeve, soft tissue treatment and electrotherapeutic modalities.

Volar dislocations of the PIP joint are uncommon and are often resistant to closed reduction. There is almost always an associated rupture of one or more collateral ligaments along with disruption of the extensor central



Figure 27.13 Splinting following a PIP joint dislocation. This dorsal block splint allows flexion but stops full extension at the PIP joint

prevention of flexion for 6 weeks and so significant loss of game time occurs in mid-season injuries. Waiting until the end of the season risks a fixed boutonnière deformity which often heals poorly with or without surgery.

Dislocations of the DIP joint

Dislocations of the DIP joint are most often due to a ball hitting the tip of the finger forcing hyperextension. They are commonly associated with a volar skin laceration. Reduction is achieved by traction and flexion. The joint is usually stable post reduction, but should still be splinted for 3 weeks in 10° of flexion.¹⁵ Controlled active flexion is permitted during this time if the joint is stable, and can be progressed gradually from that point. Collateral ligament injuries are rare. Failure to achieve reduction of the joint should be investigated, and flexor tendon function must be assessed as avulsion can occur with this injury.¹⁶

The less common volar dislocation occurs in association with a fracture and usually involves damage to the extensor tendon. This presents with the mallet finger deformity. Thus, all mallet fingers must be radiographed to exclude fracture. If volar dislocation has occurred, open repair may be indicated depending on the size and position of the fracture fragment.¹⁷

LIGAMENT AND TENDON INJURIES

Sprain of the ulnar collateral ligament of the first MCP joint

Initially known as 'gamekeeper's thumb', and more recently 'skier's thumb', this injury is one of the most common injuries to the hand and is found in all sports, especially those requiring stick holding or ball handling. Damage to the collateral ligaments of the thumb has an adverse effect

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If radiographs reveal an associated fracture of the volar lip of the middle phalanx involving more than one-third of the joint surface, open reduction and internal fixation are usually required to restore stability.

slip insertion. This injury predisposes to the development of a boutonnière (buttonhole) deformity and should be treated aggressively with either surgery or a splint, holding the PIP joint of the affected finger in extension for 6 weeks while encouraging DIP movement. Surgery requires

on the hand's strength and precision.¹⁸ This is particularly appreciated in pinching tasks such as holding a key (Fig. 27.14a).

Diagnosis is typically made clinically, with stress testing of the metacarpal joint of the thumb done in neutral and at 30° of flexion (Fig. 27.2f). Laxity in a radial direction of greater than 15° with soft end feel compared to the contralateral thumb is an indication of a complete rupture. Less than 15° indicates an incomplete tear. Radiographs are required to rule out an avulsion fracture.^{18, 19}

Surgical repair is required for complete tears because the anatomy of the ulnar aspect of the metacarpal joint can preclude opposition of the ends of the ruptured ligament. This is known as a Stener lesion for which surgical repair

is mandatory (Fig. 27.15). In general, surgical management using suture anchors with pinning of the MCP joint is preferred for full tears, with inconsistent results reported for conservative treatment.^{18, 20} However although results are better if repaired acutely, delayed repair is still successful, making the timing of surgery dependent upon factors specific to the athlete such as the sport and the time of season.²¹

Incomplete tears are splinted in a hand-based thumb spica with the thumb MCP joint placed in slight radial deviation and return to play dependent upon the athlete. The splint is worn full-time for 6 weeks, at which time a weaning and active motion protocol is initiated with strengthening at 8-10 weeks, and return to play with taping and/or a protective dorsal gutter splint for another 2-3 months (Fig. 27.14b). Postoperative rehabilitation is similar, with return to protected play as early as 2 weeks after surgery.^{19, 22}

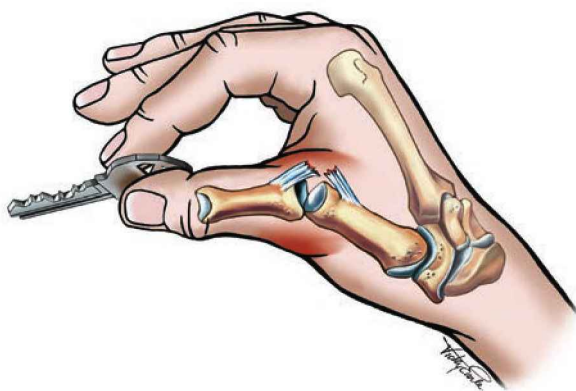
Sprain of the radial collateral ligament of the first MCP joint

Tears of the radio-collateral ligament at the MCP joint of the thumb are not as common as ulnar-sided injury and the management is mostly conservative. The major difference is that Stener-type lesions are extremely rare due to differences in anatomy. The injury occurs as a result of an ulnar directed force, be it from a punch or from a ball. Stability is usually not the chief presenting problem, with pain on grip and stiffness being the chief complaints, making delayed surgery more acceptable. Stress testing is positive with greater than 15° of laxity compared to the contralateral side and confirmation made radiologically.¹⁸

Management is usually conservative, with hand-based spica splinting in a position that minimises the pull of the adductor pollicis for 6 weeks, followed by an extended period of protected taping and splinting in play. Surgery is required for large tears with volar subluxation and chronic cases, with presentation often more delayed than for UCL injuries.^{21, 23} Immobilisation is required for 4-6 weeks afterwards, and protection in play for 2-3 months, with good long-term results reported.²⁴ Surgery is often indicated between seasons for contact sport athletes due to chronic instability.

Capsular sprain of the first MCP joint

Capsular sprains of the first MCP joint are an extremely common injury in ball-handling sports. They result from a hyperextension injury and are prone to recurrence. Treatment involves active rehabilitation and protection of the joint from hyperextension. This is achieved with the use of a thermoplastic brace over the dorsal aspect of the MCP joint.



(a)



(b)

Figure 27.14 Complete ligament tear of the first MCP joint (a) Pinching is affected (b) Protective splint worn under tape during return to sport. A hole can be created over the MCP joint to permit some flexion but still restrict deviation

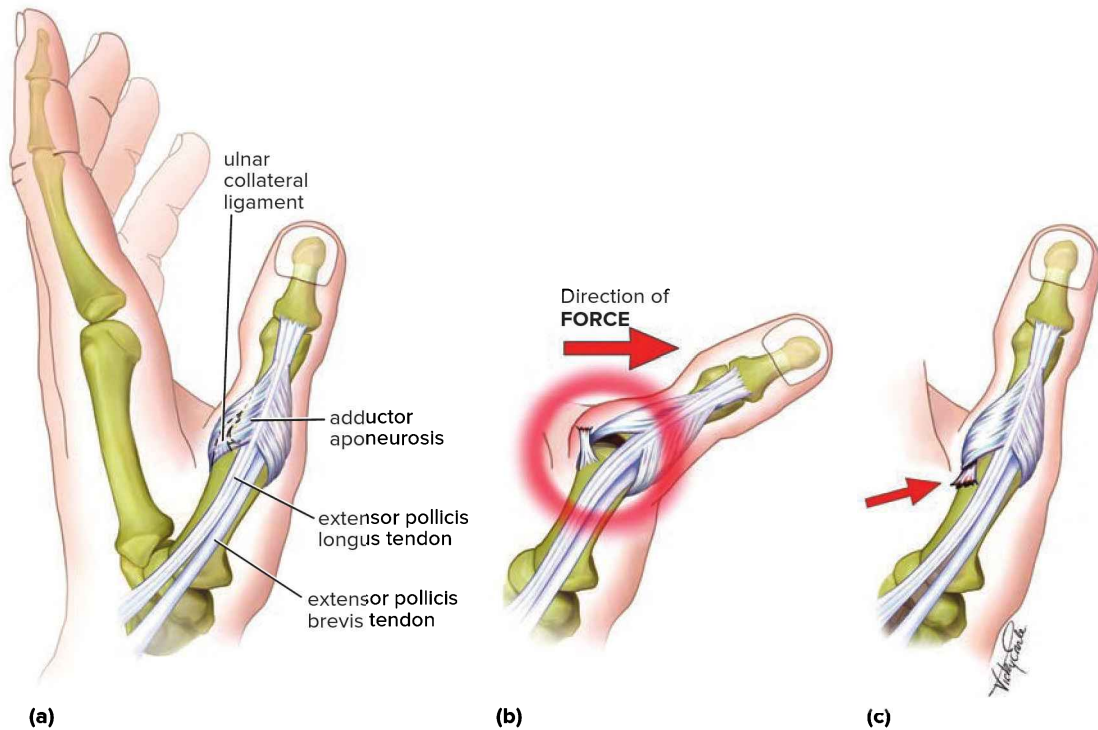


Figure 27.15 Mechanism of formation of a Stener lesion of the thumb (a) Anatomy (b) Valgus stress opens up the joint and the adductor aponeurosis slips past the distal end of the proximal portion of the ruptured ulnar collateral ligament (c) As alignment returns to normal, the adductor aponeurosis catches the proximal portion of the ulnar collateral ligament and flips it back proximally to form the Stener lesion

PIP joint sprains

The collateral ligaments of the PIP joints are commonly injured as a result of a sideways force. Partial tears of the collateral ligament are painful, but remain stable on lateral stress. Complete tears show marked instability with lateral stress. This injury also includes hyperextension stress to the volar plate, which may avulse its insertion from the base of the metacarpal.

Stable tears should be treated by 10 days of finger-based splinting with the PIP in neutral, followed by buddy taping (Fig. 27.4), swelling management and active exercises. If the joint is unstable the splint should be fashioned in 10° of flexion greater than the point of instability (Fig. 27.13). This splint should then be gradually extended towards neutral, adjusting weekly over a period of 3 weeks. Active flexion within this controlled arc is permitted and encouraged.²⁵ Complete tears of the collateral ligament should ideally be treated with surgical repair, although in most cases conservative management provides an adequate result.²⁶

Volar plate injuries without dislocation may heal normally, but also may produce a painful PIP joint or

a lax volar plate with repeated sprains and arthritis. Cortisone injected into the joint helps with chronic pain, but sometimes a volar plate arthroplasty is indicated to prevent further complications.

Combined tears of the volar plate and either collateral ligament may lead to a 'windscreen wiper' effect on the condyles of the proximal phalanx and eventual deformity. Repair is a better option.

Mallet finger

Mallet finger is a flexion deformity resulting from either avulsion of the extensor tendon from the distal phalanx or following rupture of the tendon proximal to its insertion into the distal phalanx (Fig 27.16a). As a result, it can be classified as bony or soft (Table 27.3). It affects the ulnar three digits 90% of the time and commonly results from a ball striking the extended fingertip, forcing the DIP joint into flexion while the extensor mechanism is actively contracting. This produces disruption or stretching of the extensor mechanism over the DIP joint and is often seen in baseball catchers, fielders, football receivers, cricketers and basketball players.²⁷

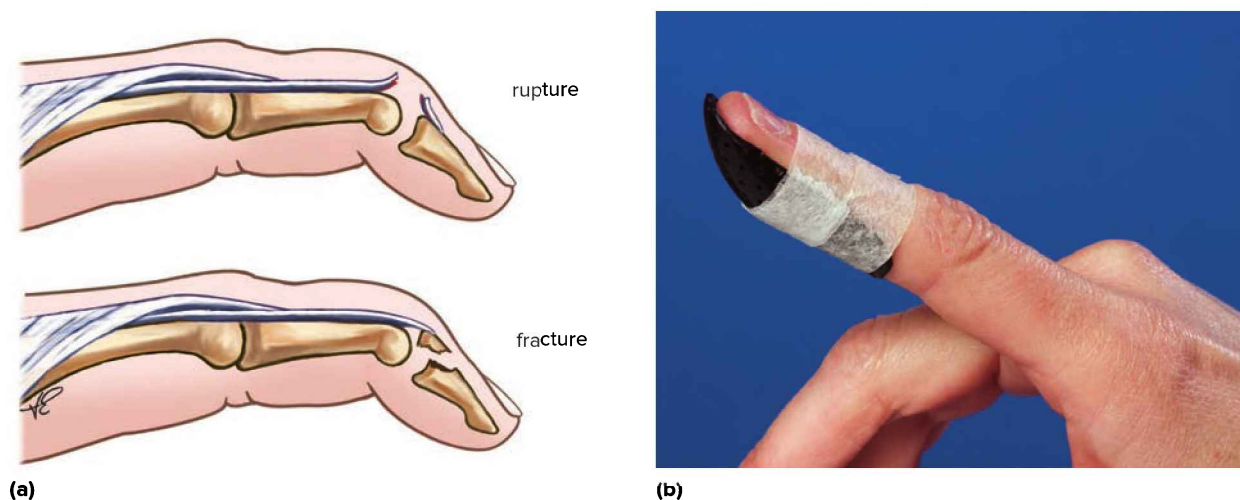


Figure 27.16 Mallet finger (a) Mechanism of deformity—rupture or avulsion (b) Stack splint. A dorsal splint is preferred for early return to sport because it will permit some tip protection and allow palmar sensation

Examination reveals tenderness over the dorsal aspect of the distal phalanx and an inability to actively extend the DIP joint from its resting flexed position. If left untreated, a chronic mallet finger type deformity develops. This flexion deformity is caused by the unopposed action of the flexor digitorum profundus tendon.

Radiography must be performed to exclude an avulsion fracture of the distal phalanx or injury and subluxation to the DIP joint. The avulsion fracture is only considered significant if greater than one-third of the joint surface is involved. Any volar subluxation requires open reduction and internal fixation to correct the joint position (the subluxation).²⁷ A fracture dislocation of the epiphyseal

plate may occur in children. Most of these injuries are type II epiphyseal injuries and closed management is preferred.

Treatment of uncomplicated mallet finger involves splinting the DIP joint in slight hyperextension for a period of 6 weeks for a bony mallet and up to 8 weeks for a soft mallet injury, with regular monitoring. The splint is then worn for an additional 4 weeks while engaging in sporting activity and at night (Table 27.3). Treatment is reinstituted at any sign of recurrence of a lag. The splint may be made of metal or plastic and applied to either the volar (Fig. 27.16b) or the dorsal surface. Patients with dorsal splints maintain pulp sensation, which may

Table 27.3 Mallet finger treatment protocol

	Type of injury	
	Bony mallet	Soft mallet
Continuous splinting	Thermoplastic palmar splint for 6 weeks (2/52 more if lag)	Thermoplastic palmar splint for 8 weeks (2/52 more if lag)
Weaning protocol		
Week 1 after continuous splinting period over	Off for 1 hour am, 1 hour pm On all other times	As for bony
Week 2	Off for 2 hours am, 2 hours pm On all other times	As for bony
Week 3	Off for 4 hours am, 4 hours pm On all other times	As for bony
Week 4	Off during day, on during sleep and sport	As for bony
Weeks 5 and 6	Tape for sport. Initiate exercises to increase DIP flexion if required	Tape for sport

be more suitable should the athlete insist on returning to sports before the injury has healed. The finger should be kept dry and examined regularly for skin slough and maceration.²⁸

When treating mallet finger, emphasise to the patient that the joint must be kept in hyperextension at all times during the first 6–8 weeks, even when the splint is removed for cleaning. If a patient is not prepared to do this, then surgery should be considered. The consequences of not splinting are a chronic mallet finger type flexion deformity with osteophyte formation and degeneration of the DIP surface, and hyperextension at the PIP joint.

Chronic swan neck deformity

There is commonly confusion between the presentation of swan neck and boutonnière deformities (Fig. 27.17). Hyperextension of the PIP with DIP joint flexion is characterised as a swan neck deformity. It can be a sequela to mallet finger injury or volar plate disruption.²⁸ Chronic cases will demonstrate instability of the PIP joint with visible snapping and discomfort. Fine motor activities become difficult and the finger can lock. Given that palliative treatment can be provided with dorsal blocking, game-day taping or functional figure-eight

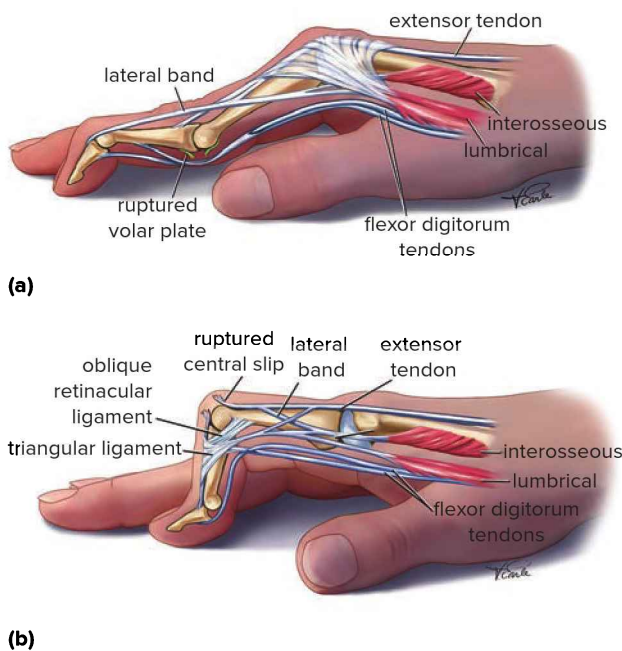


Figure 27.17 There is commonly confusion between the presentation of (a) swan neck and (b) boutonnière deformities



Figure 27.18 Functional figure-eight splint for swan neck deformity preventing hyperextension but permitting PIP flexion

splints (Fig. 27.18),²⁹ many athletes opt to delay surgical reconstruction to the end of their season or career.

Boutonnière deformity

Injury to the extensor tendon of the finger just proximal to its insertion on the middle phalanx can result in a boutonnière deformity, characterised by PIP joint flexion and DIP joint hyperextension. It is a common injury in ball handling sports like basketball, football and volleyball.^{30, 31} With the central slip damaged, the lateral bands of the extensor mechanism migrate volarly and proximally, flexing the PIP joint and extending the DIP joint. This is then exacerbated by the now unopposed flexor digitorum superficialis. Without correction, ligaments will contract and the position become entrenched (Fig. 27.19).³²

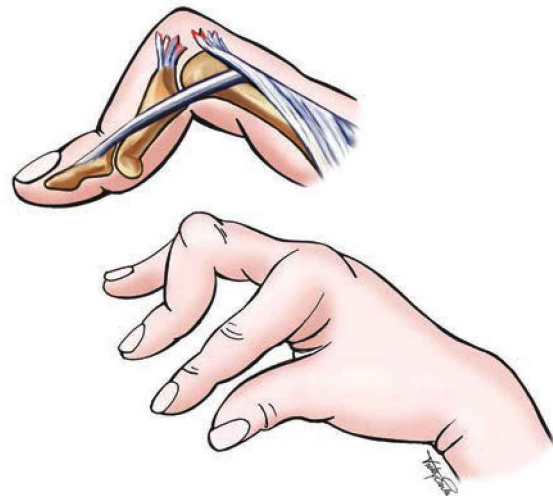


Figure 27.19 Boutonnière deformity

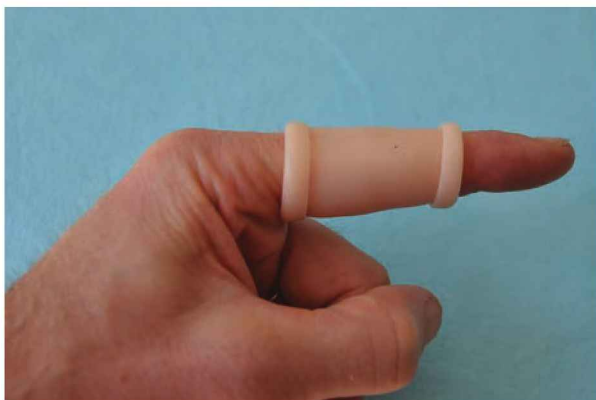


Figure 27.20 Barrel splint holding the PIP in extension but permitting DIP flexion

The deformity may not be immediately obvious and can take several weeks to develop, making an early accurate assessment important, but also highlighting the need for ongoing evaluation where injury is suspected.^{32, 33} Typically there will have been blunt trauma to the dorsum of the PIP or an instance of PIP joint hyperflexion. Open laceration requires immediate surgical consult and intervention. There will usually be significant swelling, and radiographs may show an avulsion fracture. The central slip can be tested by hyperextending the MCP joint and then asking the athlete to actively extend the PIP joint. If this can be done fully, the central slip is intact.³²

Management of the closed injury involves extension splinting of the PIP joint continuously for 6–8 weeks, encouraging regular active DIP joint flexion (Fig. 27.20). Ongoing night splinting may be required. Controlled active PIP joint flexion protocols introduced after 3 weeks of PIP joint extension have been shown to have fewer complications but require consistent monitoring, and are best instituted by a skilled hand therapist.^{33, 34}

Surgical repair of open or chronic injuries often requires reconstruction and/or pinning of the PIP joint, which can preclude a return to sport for 6 weeks until the pin is removed. Reconstruction of a chronic injury is more successful if a flexion contracture has been reduced to a more neutral position with a splint.³² For the athlete with a closed boutonnière, if they can play with a splint that maintains PIP extension, then immediate return is possible.^{30, 31}

Avulsion of the flexor digitorum profundus tendon ('jersey finger')

This injury is most commonly seen in the ring finger and may be caused by the player grabbing an opponent's clothing, resulting in the distal phalanx being forcibly extended while the athlete is actively flexing. The patient

often feels a 'snap'. The condition is often referred to as 'jersey finger'.

Examination may reveal the finger assuming a position of extension relative to the other fingers. There is an inability to actively flex the DIP joint of the affected finger. Radiography should be performed to exclude an associated avulsion fracture of the distal phalanx. A bone fragment may be seen volar to the middle phalanx or PIP joint. A lump may be palpated more proximally in the finger corresponding to the avulsed tendon.

PRACTICE PEARL

Treatment is urgent surgical repair with reattachment of the profundus tendon to the distal phalanx. This must take place within 10 days of the injury.

Late treatment requires the use of a two-stage operation, with a silicon rod being placed for some months in the finger to create the tendon sheath gap for the tendon graft to slide in. This is sometimes considered too big an operative insult, and fusing the DIP joint without grafting the tendon gives a nearly acceptable result.

Lacerations and infections of the hand

Lacerations to the fingers and hand occur frequently in sport as a result of contact with equipment such as the undersurface of a football boot. All lacerations have the potential to become infected and should, therefore, be thoroughly cleaned with an antiseptic solution and observed closely for signs of infection. Tetanus toxoid should also be administered where appropriate.

A particular concern is a laceration of the hand, often over the MCP/PIP joint, caused by teeth, usually from a punch to the mouth. These injuries should always be assumed to be contaminated and an immediate course of a broad-spectrum antibiotic should be commenced. The wound should not be closed.

Lacerations over volar DIP or PIP joints may represent compound dislocations. If this has occurred, the joint has been contaminated and the patient requires hospital admission for surgical debridement and repair. Otherwise septic arthritis may follow.

OVERUSE CONDITIONS OF THE HAND AND FINGERS

Important but sometimes overlooked are the overuse problems associated with the hand. These include trigger finger, flexor pulley injuries and other small joint injuries that are commonly seen in rock climbers.^{35–37} Trigger finger is caused by a tenosynovitis in the flexor tendon

that is large enough to be impeded by the proximal A1 (annular) pulley located at the base of the finger. Diagnosis is made clinically, with the patient presenting with often painful clicking and locking of the finger as it actively flexes. Ultrasound is used for confirmation. Conservative treatment involves splinting to prevent metacarpal flexion only, and local treatment to reduce the enlarged tendon. Corticosteroid injection is often advocated first, and then surgical release of the impeded A1 pulley.^{38, 39}

Rock climbers put chronic severe stresses on the flexor tendons and can damage or tear the important A2 and/or A4 pulleys, producing bowstringing. This often appears to be simple swelling initially, but can become quite severe. Most serious climbers wear tape or circular splints to control the tension, but surgical reconstruction may be required.

SURGICAL REFERRALS FOLLOWING HAND AND FINGER INJURY

There are a number of conditions that require urgent referral to a hand surgeon. These are shown in Table 27.4.

Exercises for the hand

Exercises following hand injury should be prescribed so as to include both intrinsic and extrinsic hand musculature.

The tendon gliding sequence of exercises does this well, and passive stretches that push into these positions should also be encouraged when appropriate in the case of hand and finger stiffness (Fig. 27.21). Strengthening exercises utilising Theraputty®, grippers and stress balls are useful once the injured finger is stable but should not be introduced before joint or fracture stability is achieved. Building up the handles on a barbell or chin-up bar will increase grip strength demands, weight plate pinches will isolate intrinsic muscles, and finger extensors and the interossei can be worked using rubber bands.

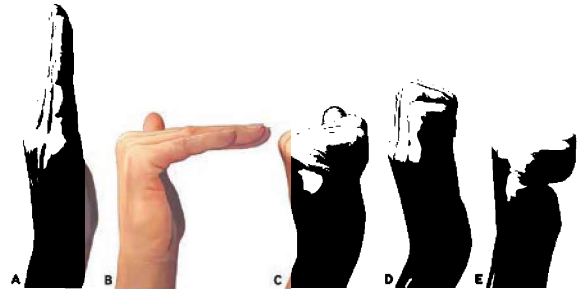


Figure 27.21 Tendon gliding sequence of exercises

Table 27.4 Urgent surgical referrals for sports hand conditions

Injury	Common sports	Management notes
Flexor digitorum profundus avulsion	Tackling sports, including rugby and all football codes	Early: acute surgical repair Late (>6 weeks): two-stage reconstruction
All compound lacerations	Contact sports and/or motor sports and/or cycling i.e. high velocity sports	Surgical washout (local or general) Explore to exclude foreign bodies (especially teeth, soil) and debride
Sagittal band rupture MCPs 2,3,4	Boxing	Sagittal band reconstruction (not as urgent)
Mallet fracture/dislocation	Ball sports	Joint reduction and pinning
Ulnar collateral ligament rupture, MCP thumb	Skiing, football ('thumb plant')	Splint if incomplete Surgical reconstruction if Stener lesion
Bennett's/Rolando fracture base of 1st MC	Contact sports	Must reduce joint surface for good result
Rotated fracture, middle phalanx	Contact sports	Does not tolerate shortening or rotation
Rotated fracture, proximal phalanx	Contact sports	Often leaves stiff PIP if left unreduced
Rotated metacarpal fracture	Contact sports	Causes crossover of fingers; quicker return in professional sports
CMC subluxation and dislocation	Boxing, martial arts	Must be reduced to maintain strength and power; if late needs CMC fusion

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e